

LECTURE-8

□ Single Cycle. Data Path: How data is flowing

⇒ Basic Instr. Types:

① Instruction Memory (PC) : Reads each instructions one by one to decode. It has a PC for which is has an ALU for incrementing PC ($PC = PC + 4$)

② R-Type: $\overset{rd}{add} \overset{rs1}{x5}, \overset{rs2}{x2}, x3$

Instruction read $\rightarrow$ Two values reads & sent to $rs1$ & $rs2$
 Do specific arithmetic operation with func 7, $\leftarrow$
 func 3 and opcode; ALU will perform operation (add).
 $\rightarrow$ Store result in "rd" ($x5$)

Load: Address in $RS1$.

$$RS1 + \text{offset} = \text{address}$$

- go to that address and read the data
- Load that data into 'rd'

Store: Calculate address ($rs1 + \text{offset}$)

- go to that address and write the data given in 'rs2'

$\Rightarrow$ Branching (if conditions): $beq, x5, x6, \text{label_BTA}$

Branch (executed or not)

$\leftarrow$ not

$\rightarrow$ executed

(next line) $PC = PC + 4$

$PC = PC + \text{BTA}$ (branch target address)

*More than one path/links of data means always use MUX.

Diagrams from slides.